CPSC 321: Homework 6 (Normalization) *Due: Tuesday, Nov. 10, 2026*

Instructions. Your answers to the following questions must be typed (not handwritten), organized, and clearly marked (i.e., with respect to which question and part is being answered). Additionally, include your first and last name, the course (CPSC-321, Fall 2026), and the assignment (HW-6). Your work must be submitted as a single PDF file called `hw-6.pdf` via Canvas by midnight at the end of the due date. This assignment is worth 15 points. Note that this homework will likely take you a bit longer than you might think, and you will need to start early to ensure you have time to ask questions (and receive answers) before the due date.

Assumptions. Assume atomic values and no `NULL`s. For Questions 3–6, base your answers on the stated functional dependencies. Do not assume additional dependencies based on the attribute names. As in class, XY means the union of attribute sets X and Y.

1. [2 pts] Give an example of a relation (different from what we have done in class and in the reader) that contains a primary key along with two non-key, non-trivial functional dependencies. (a) State the relation, its primary key, and the functional dependencies, and briefly explain what each dependency means in your example; (b) give an instance with at least three rows that satisfies the dependencies and contains redundant data; and (c) give an example of each type of anomaly that can occur (update, insertion, and deletion). For each anomaly, describe the operation and the resulting problem using your example.

2. [2 pts] Given a set of functional dependencies (FDs), we can use the following rules (called *Armstrong's Axioms*) to infer new dependencies implied by the set.

- *Reflexivity*: If $Y \subseteq X$, then $X \to Y$
- *Augmentation*: If $X \to Y$, then $XZ \to YZ$ for any Z
- *Transitivity*: If $X \to Y$ and $Y \to Z$, then $X \to Z$

Use Armstrong's three axioms to provide direct proofs of the following two rules. Number your steps and state the axiom and the given or previously derived FDs used at each step.

(a). *Decomposition*: If $X \to YZ$, then $X \to Y$ and $X \to Z$.

(b). *Pseudotransitivity*: If $X \to Y$ and $YW \to Z$, then $XW \to Z$.

3. [3 pts] FDs can be used to determine a relation's candidate keys. In particular, a candidate key is a minimal set of attributes that all other attributes functionally depend on. We can test if a set of attributes is a superkey by repeatedly applying Armstrong's axioms and/or the rules in question 2. Find *all of the candidate keys* for each of the following scenarios. For full credit, you *must* explain why each key determines every attribute, why it is minimal, and why there are no other candidate keys.

(a). A relation $R(a, b, c, d)$ and FDs $\{b \to c, d \to a\}$.

(b). A relation $R(a, b, c, d, e)$ and FDs $\{a \to bc, cd \to e, b \to d, e \to a\}$.

(c). A relation $R(a, b, c, d, e)$ and FDs $\{c \to b, bd \to e, a \to d, e \to a\}$.

4. [3 pts] Assume we have the following (at least 1NF) relational schema:

`movie_info(title, director, studio, studio_loc, award_type, award_year)`

with functional dependencies

`award_type, award_year` $\rightarrow$ `title, director`

`studio` $\rightarrow$ `studio_loc`

`director` $\rightarrow$ `studio`

For each of the following decomposed `movie_info` tables, state the candidate key(s) for the table and the highest normal form each table is in (i.e., either 1NF, 2NF, 3NF, or BCNF). Treat each part independently. Use the given FDs to explain why each table satisfies the normal form you give. If it is not in BCNF, identify an FD that prevents it from satisfying the next higher normal form.

(a). `movie_info′(award_type, award_year, director, studio_loc)`

(b). `movie_info′(title, director, studio, studio_loc)`

(c). `movie_info′(award_type, award_year, studio_loc)`

(d). For the original relation $R(a, b, c, d, e)$ from Question 3(b), what is its highest normal form? Explain your answer using the candidate keys you found and the given FDs.

5. [2 pts] Consider the relation $R(a, b, c, d, e)$ with FDs $\{a \rightarrow bc, cd \rightarrow e, b \rightarrow d, e \rightarrow a\}$ and the decomposition $R_1(a, b, c)$ and $R_2(a, d, e)$.

(a). Is the decomposition lossless? Justify your answer using the attributes shared by R_1 and R_2 and the given FDs.

(b). Is the decomposition dependency preserving? Justify your answer.

6. [3 pts] Give a lossless and dependency preserving decomposition of the full (original) `movie_info` relation in Question 4 using the "BCNF Algorithm" from class. Draw the decomposition as a "tree", as we've done in class, to show your work. Identify the FD used for each split and include the candidate keys of the final decomposed relations. Briefly explain why the final relations are in BCNF, why the decomposition is lossless, and why it preserves the original dependencies.